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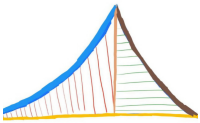

G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018
First edition published on July 10, 2024
In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1 Signal Processing

1.1 Analog

1.1.1 A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007)

a) 2 Ns/m b) 2 N/s c) 0.2 kg/s d) 0.2 N/m

1.1.2 A 1 kg mass attached to a spring elongates it by 16 mm. The mass is then pulled from its equilibrium position by 10 mm and released from rest. Assuming the acceleration due to gravity of $9.81 m/s^2$, the response of the mass in mm is given by: (AE 2007)

a) $x = \cos 4.76\,t$ b) $x = \sin 4.76\,t$ c) $x = \sin 16\,t$ d) $x = 10 \cos 16\,t$

1.1.3 A spring-mass-damper system is excited by a force $F_0 \sin \omega t$. The amplitude at resonance is measured to be 1 cm. At half the resonance frequency, the amplitude is 0.5 cm. The damping ratio of the system is (AE 2007)

a) 0.1026 b) 0.3242 c) 0.7211 d) 0.1936

1.1.4 $\lim_{x \to 0} \frac{\sin x}{e^4 x} =$ (AE 2007)

a) 10 b) 0 c) 1 d) ∞

1.1.5 Let a dynamical system be described by the differential equation $2\frac{dx}{dt} + \cos x = 0$. Which of the following differential equations describes this system in a close approximation sense for small perturbation about $x = \pi/4$? (AE 2007)

a) $2\frac{dx}{dt} + \sin x = 0$
b) $2\frac{dx}{dt} - \frac{1}{\sqrt{2}} x = 0$
c) $\frac{dx}{dt} + \cos x = 0$
d) $\frac{dx}{dt} + x = 0$

1.1.6 Consider the spring mass system shown in the figure below. This system has two degrees of freedom representing the motions of the two masses. (AE 2007)

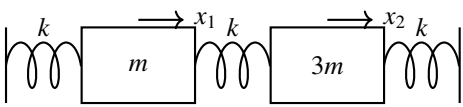

Fig. 1.1.6.1

a) The system shows the following type of coordinate coupling
   i) Static coupling
   ii) Dynamic coupling
   iii) Static and dynamic coupling
   iv) No coupling
b) The two natural frequencies of the system are given as

i) $\sqrt{\frac{4\pm\sqrt{5}k}{3m}}$ ii) $\sqrt{\frac{4\pm\sqrt{3}k}{3m}}$ iii) $\sqrt{\frac{4\pm\sqrt{7}k}{3m}}$ iv) $\sqrt{\frac{4\pm\sqrt{11}k}{3m}}$

1.1.7 Let

$$F(s) = \frac{(s+10)}{(s+2)(s+20)}$$

(AE 2007)

a) The partial fraction expansion of $F(s)$ is

i) $\frac{1}{s+2} + \frac{1}{s+20}$ ii) $\frac{5}{s+2} + \frac{2}{s+20}$ iii) $\frac{2}{s+2} + \frac{20}{s+20}$ iv) $\frac{4/9}{s+2} + \frac{5/9}{s+20}$

b) The inverse Laplace transform of $F(s)$ is

i) $2e^{-34} + 20e^{-34}$
ii) $\frac{4}{9}e^{-34} + \frac{5}{9}e^{-34}$
iii) $5e^{-34} + 2e^{-34}$
iv) $\frac{9}{4}e^{-34} + \frac{9}{5}e^{-34}$

1.1.8

$$\frac{\omega}{(s+a)^2+\omega^2}$$

is the Laplace Transform of (AG 2007)

a) $\exp(-at)\sin\omega t$
b) $\sin\omega t$
c) $\exp(-at)\sinh\omega t$
d) $\sinh\omega t$

1.1.9 Evaluate $\int_0^{\pi} \frac{\sin t}{t} dt$ (CE 2007)

a) π b) $\pi/2$ c) $\pi/4$ d) $\pi/8$

1.2 Discrete

1.2.1 The Euler iteration formula for numerically integrating a first order nonlinear differential equation of form $x = f(x)$, with a constant step size of Δt is (AE 2007)

a) $x_{k+1} = x_k - \Delta t \times f(x_k)$
b) $x_{k+1} = x_k + \left(\Delta t^2/2\right) \times f(x_k)$
c) $x_{k+1} = x_k - (1/\Delta t) \times f(x_k)$
d) $x_{k+1} = x_k + \Delta t \times f(x_k)$

1.2.2 Numerical values of the integral $\int_0^1 \frac{1}{1+x^2} dx$ if evaluated numerically using the Trapazoidal rule with dx = 0.2 would be (AE 2007)

a) 1 b) $\pi/4$ c) 0.7837 d) 0.2536

1.2.3 The Newton-Raphson iteration formula to find a cube root of a positive number c is (AE 2007)

a) $x_{k+1} = \frac{2x_k^3 + \sqrt[3]{c}}{3x_k^2}$ b) $x_{k+1} = \frac{2x_k^3 - \sqrt[3]{c}}{-3x_k^2}$ c) $x_{k+1} = \frac{2x_k^3 + c}{3x_k^2}$ d) $x_{k+1} = \frac{x_k^3 + c}{3x_k^2}$

1.2.4 $y(0) = 0$ and using Euler's method with step size $h = 0.1$ solution of the differential equation

$$\frac{dy}{dx} = 2xy + 1$$

gives the value of $y(0.3) =$? (AG 2007)

a) 0.3101 b) 0.3142 c) 0.6202 d) 4.080

1.2.5 Integrating the function (AG 2007)

$$f(x) = 1 + e^{-x}\sin(4x)$$

over the interval [0, 1] using Simpson's $\frac{1}{3}$ rule gives Result = ?

a) 1.021 b) 1.091 c) 1.321 d) 2.642

1.2.6 The degree of the differential equation $\frac{d^2x}{dt^2} + 2x^3 = 0$ is (CE 2007)

a) 0 b) 1 c) 2 d) 3

1.2.7 The solution of the differential equation $\frac{dy}{dx} = x^2y$ with the condition that $y = 1$ at $x = 0$ is (CE 2007)

a) $y = e^{\frac{1}{2x}}$ b) $\ln(y) = \frac{x^3}{3} + 4$ c) $\ln(y) = \frac{x^2}{2}$ d) $y = e^{\frac{x^3}{3}}$

1.2.8 The following equation needs to be numerically solved using the Newton-Raphson method. $x^3 + 4x - 9 = 0$ The iterative equation for this purpose is (k indicates the iteration level) (CE 2007)

a) $x_{k+1} = \frac{2x_k^3+9}{3x_k^2+4}$

b) $x_{k+1} = \frac{2x_k^3+9}{3x_k^2+9}$

c) $x_{k+1} = x_k - \frac{x_k^3+4x_k-9}{3x_k^2+4}$

d) $x_{k+1} = \frac{4x_k^2+3}{9x_k^2+2}$

1.2.9 Given that one root of the equation $x^3 - 10x^2 + 31x - 30 = 0$ is 5, the other two roots are (CE 2007)

a) 2 and 3 b) 2 and 4 c) 3 and 4 d) -2 and -3

1.2.10 Consider the series $x_{n+1} = \frac{x_n}{2} + \frac{9}{8x_n}, x_0 = 0.5$ obtained from the Newton-Raphson method. The series converges to (CS 2007)

a) 1.5 b) $\sqrt{2}$ c) 1.6 d) 1.4

2 Optimization

2.0.1 The minimum value of

$$J(x) = x^2 - 7x + 30$$

occurs at (AE 2007)

a) $x = 7/2$ b) $x = 7/30$ c) $x = 30/7$ d) $x = 30$

3 Probability

3.0.1 For station X, the maximum one day rainfall with 25 years return period is 100 mm. The probability of a one day rainfall equal to or greater than 100 mm at station X occurring at least once in 15 successive years is (AG 2007)

a) 0.458 b) 0.500 c) 0.637 d) 0.990

3.0.2 If the standard deviation of the spot speed of vehicles in a highway is $8.8kmph$ and the mean speed of the vehicles is $33kmph$, the coefficient of variation in speed is (CE 2007)

a) 0.1517 b) 0.1867 c) 0.2666 d) 0.3646

3.0.3 Suppose we uniformly and randomly select a permutation from the 20! permutations of $1, 2, 3, ..., 20$. What is the probability that 2 appears at an earlier position than any other even number in selected permutation ? (CS 2007)

a) $\frac{1}{2}$ b) $\frac{1}{10}$ c) $\frac{9!}{20!}$ d) None of these

3.0.4 There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot? (CS 2007)

a) $np(1-p)^{n-1}$ b) $(1-p)^{n-1}$ c) $p(1-p)^{n-1}$ d) $1-(1-p)^{n-1}$